# Topic 1 Review

Algebra I · Module 3, Topic 1: Linear Equations and Inequalities · Tuesday, Nov 17 · Test Wed Nov 18

Name _____ Period _____ Date _____

**Directions:** This review has the same kinds of questions as the Topic 1 test, in the same order. Show your work. For multiple choice, choose one answer. Finish Part 1 (just like the test) first; Part 2 is extra practice.

## Part 1 · Just like the test (Questions 1–11)

**1.** Jayden has $40 in his bank account. He deposits $15 each week to save for new headphones. Which equation can be used to determine the amount that Jayden has saved, $y$, after $x$ weeks?

**A.** $y = 15 + 40x$

**B.** $y = 15x + 40$

**C.** $y = -15x + 40$

**D.** $y = -40x + 15$

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**2.** What value of $k$ makes the equation $5 - 4(k + 3) = 9 - 2(2k + 6)$ true?

**F.** 1

**G.** 2

**H.** 4

**J.** No solution

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**3.** What is the solution set for $3(4 - t) > -2t + 15$?

**A.** $t > -3$

**B.** $t < 3$

**C.** $t \geq 3$

**D.** $t < -3$

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**4.** Solve the equation shown.

$0.3(4x - 6) = -1.8x + 4.2$

Answer: _____

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**5.** Does the equation shown below have one solution, no solution, or an infinite number of solutions? Explain your reasoning.

$\frac{1}{3}(12x + 30) = 4(x + 2) + 2$

Answer: _____

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**6.** Maya goes to a school store. She buys 4 notebooks that cost $1.75 each and some pens for $1.20 each. Write an equation for the total cost, $y$, when Maya buys $x$ pens.

Answer: _____

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**7.** Rewrite the equation in slope-intercept form.
$2x + 5y = 15$

Answer: _____

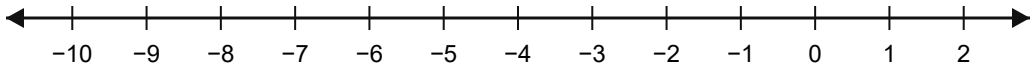
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**8.** The formula for the volume of a rectangular prism is $V = \ell wh$. Solve the equation for $h$.

Answer: _____

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**9.** Solve the inequality, and graph the solution on the number line.
$2.5m - 3 < -1.5(m + 18)$




Answer: _____

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**10.** Rewrite the equation $y - 4 = 3(x + 2)$ in slope-intercept form.

Answer: _____

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**11.** A triangle has a height of $(3n - 6)$ inches and a base of 10 inches. A rectangle has a length of 5 inches and a width of $(n + 4)$ inches. The area of the triangle is equal to the area of the rectangle. What is the value of $n$?

Answer: _____

## Part 2 · Extra practice (Questions 12–15)

**12.** Which equation has **no solution**?

**F.** $3(x - 2) = 3x + 5$

**G.** $3(x - 2) = 3x - 6$

**H.** $3(x - 2) = 2x + 5$

**J.** $3x - 2 = 2x - 3$

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**13.** What is the solution set for $-4x + 7 \geq 19$?

**A.** $x \geq -3$

**B.** $x \leq 3$

**C.** $x \leq -3$

**D.** $x \geq -6.5$

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**14.** The perimeter of a rectangle is $P = 2\ell + 2w$. Solve for $w$.

Answer: _____

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**15.** A gym charges a \$25 sign-up fee plus \$8 per class. Lena can spend at most \$90. Write and solve an inequality for the number of classes, $c$, she can take.

Answer: _____

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